Mini Presentation

clay and melon dataset

Presented By

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DATASET 1: CLAY

What is this experiment about?

Scientist wanted to know if adding more clay to soil could reduce the growth of a harmful bacteria called Ralstonia. They tested **11 different clay percentages (4%, 6%, 8%, 10%, 12%, 14%, 20%, 22%, 24%, 28%, 32%)** and measured bacteria levels at **6 different time points (day 2, 15, 29, 43, 58, 73).**

```
'data.frame':   69 obs. of  3 variables:
 $ per.clay : int  4 4 4 4 4 4 32 32 32 32 ...
 $ days     : int  2 15 29 43 58 73 2 15 29 43 ...
 $ ralstonia: num  7.3 6.99 6.31 5.97 5.54 ...
    per.clay          days         ralstonia
 Min.   : 4.00   Min.   : 2.00   Min.   :0.457
 1st Qu.: 8.00   1st Qu.:15.00   1st Qu.:5.657
 Median :14.00   Median :29.00   Median :6.478
 Mean   :15.77   Mean   :34.09   Mean   :5.809
 3rd Qu.:24.00   3rd Qu.:58.00   3rd Qu.:6.992
 Max.   :32.00   Max.   :73.00   Max.   :7.591
```

Summary:

The clay data set contains **69 measurements** of Ralstonia bacteria levels under different soil clay percentages and observation days. The variable **per.clay** represents the **treatment factor,** while **days** represents a **blocking factor** accounting for **time variation**. The **response variable ralstonia** measures **bacterial growth levels**

Why was days chosen as a blocking factor and not a treatment factor?

Because the researcher wasn't interested in studying the effect of time itself — they wanted to study clay percentage. **Time was included only to control for the natural change in bacteria levels over time.** If we ignored days, that time variation would end up in the error term and make it harder to detect the real clay treatment effect. A **blocking factor absorbs unwanted variation,** it's not what we're testing.

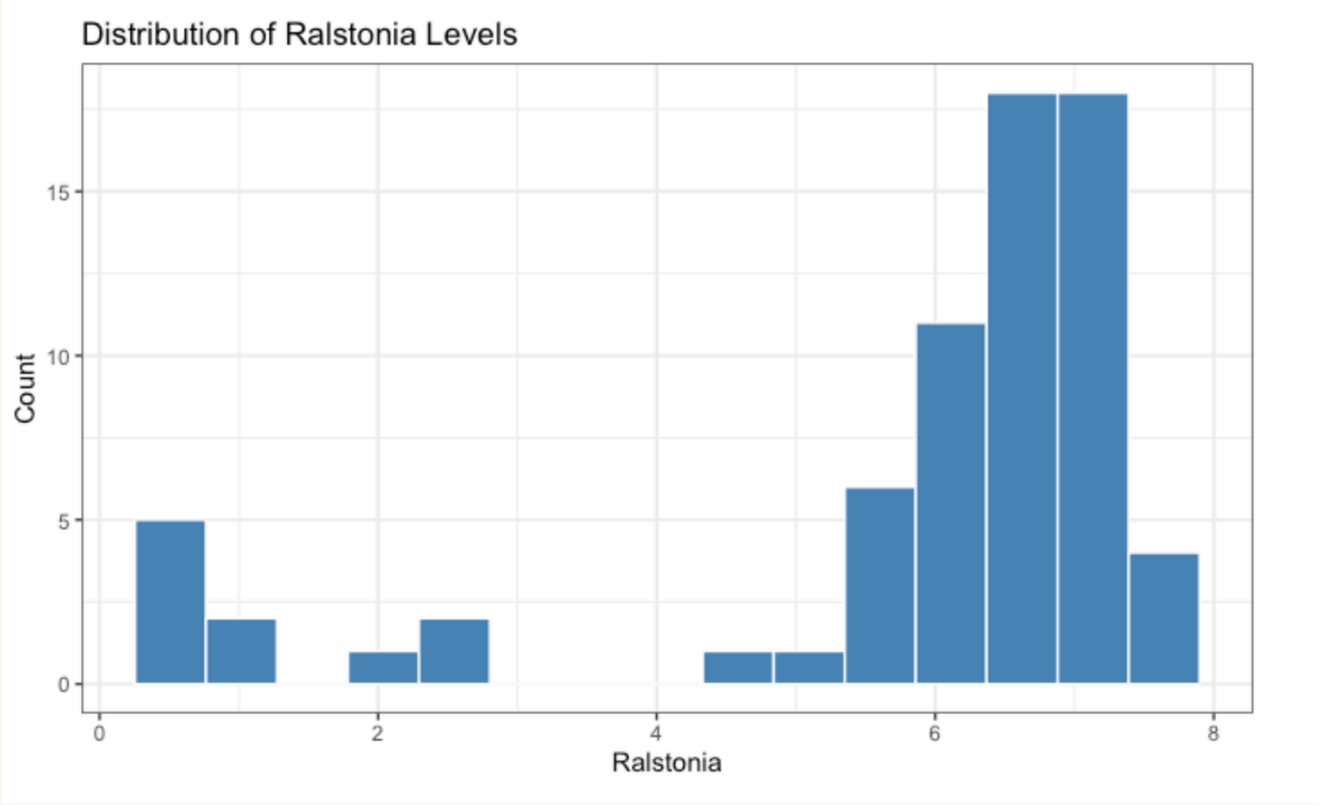
What are the variables?

per.clay — this is what is being tested. Think of it as the ingredient being changed in each experiment. 11 different amounts of clay were used.

days — Scientist measured bacteria at different time points (day 2, 15, 29, 43, 58, 73). They included this because bacteria levels naturally change over time and they didn't want time to mess up the results.

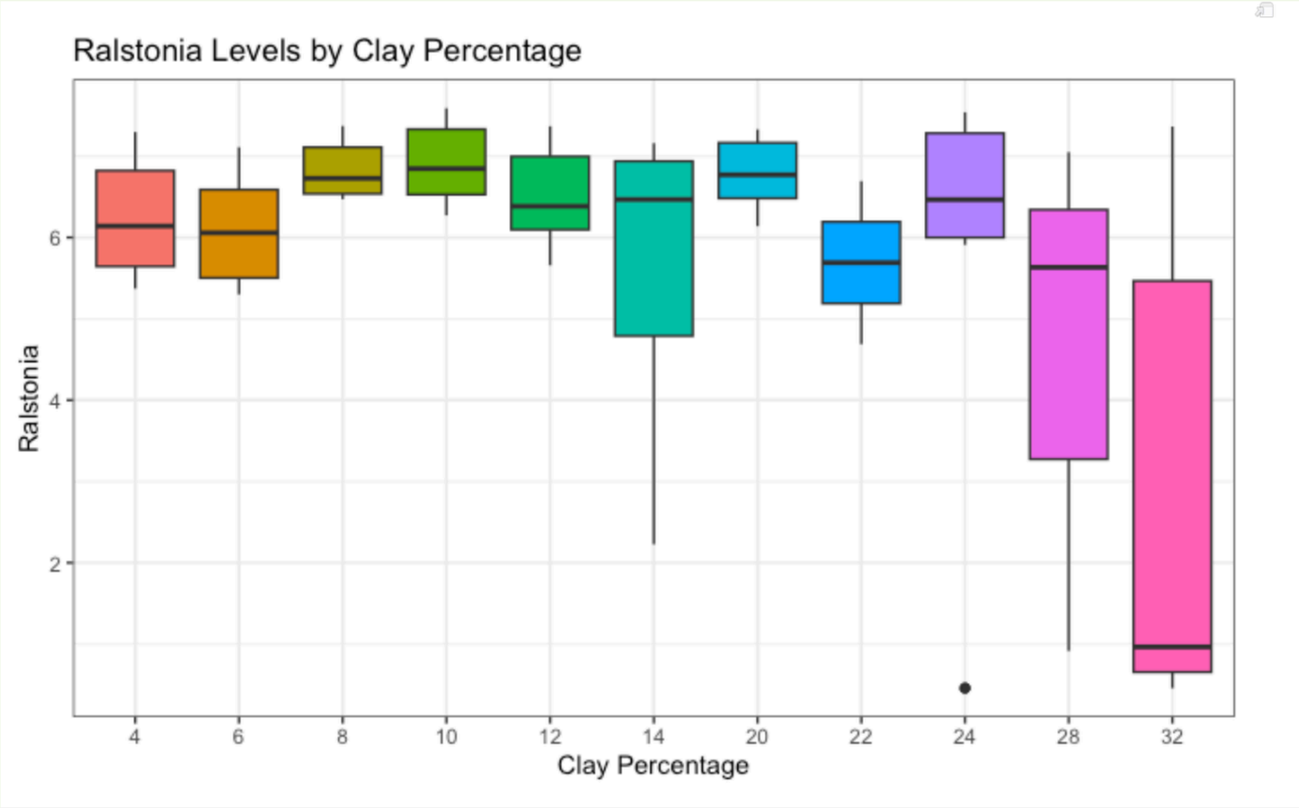
ralstonia — this is what they measured. How much bacteria is present. Higher number = more bacteria.

Looking at the Data: Plot 1 — Histogram



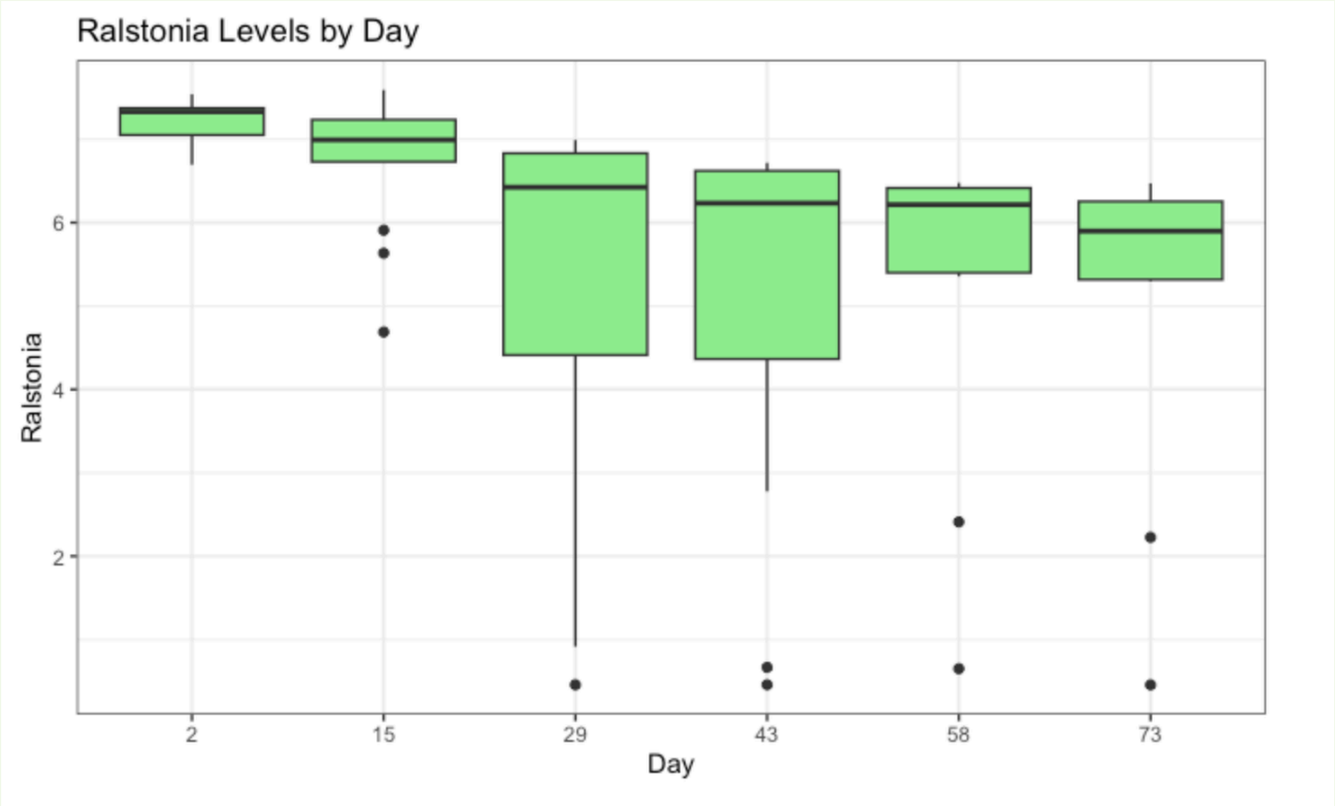
Shows the overall distribution of Ralstonia levels across the entire experiment. This gives us a first look at whether the data is roughly symmetric or skewed.

Looking at the Data: Plot 2 — Boxplot by clay percentage



Shows how Ralstonia levels vary across the 11 clay treatments. We can already see that higher clay percentages, particularly 32%, appear to have lower median Ralstonia levels compared to lower clay percentages.

Looking at the Data: Plot 3 — Boxplot by day



Shows how Ralstonia levels change over time. We can see a clear downward trend as days increase, particularly after day 15. This visually justifies including days as a blocking factor since bacteria levels clearly change over time.

ANOVA: Two-Way

```{r}
clay$per.clay <- as.factor(clay$per.clay)
clay$days <- as.factor(clay$days)

model_clay <- aov(ralstonia ~ per.clay + days, data = clay)
summary.aov(model_clay)
cv.model(model_clay)
```

```
            Df Sum Sq Mean Sq F value   Pr(>F)
per.clay    10  81.65   8.165    3.45 0.001524 **
days         5  70.88  14.176    5.99 0.000184 ***
Residuals   53 125.44   2.367
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
[1] 26.48258
```

$Y_{ij} = \mu + \alpha_i + \beta_j + \varepsilon_{ij}$

$ralstonia_{ij} = \mu + \alpha_i + \beta_j + \varepsilon_{ij}$

- **$ralstonia_{ij}$** = the observed Ralstonia level for the ith clay treatment in the jth day block
- **μ** = the overall grand mean of all Ralstonia measurements
- **α_i** = the effect of the ith clay treatment *(i = 4, 6, 8, 10, 12, 14, 20, 22, 24, 28, 32)*
- **β_j** = the effect of the jth day block *(j = 2, 15, 29, 43, 58, 73)*
- **ε_{ij}** = the random error term for each observation

ralstonia = grand mean + effect of per.clay + effect of days + random error (ε)

The residual mean square is 2.367 — this is the size of our random error term, ε ,epsilon. The smaller this is the better the model fits the data

ANOVA: Two-Way

Does changing the clay percentage actually change the bacteria levels, or are any differences just random chance?

days:
The blocking factor, days, is highly significant *(F = 5.99, df = 5, p = 0.000184)*. Therefore, time significantly affect bacteria levels, as it changes over time. This justifies the use of days as a blocking factor — it absorbed time-related variability and improved the precision of treatment comparisons.

per.clay:
The ANOVA test reveals a significant effect of clay percentage on Ralstonia levels *(F = 3.45, df = 10, p = 0.0015)*. This means at least one clay percentage produces a significantly different bacterial level compared to the others.

Conclusion:
These results indicate that both soil composition and time influence Ralstonia growth.

Cross-Validation:
The CV% = 26.48%. This just means there was moderate variability in the experiment. Basically the data was a little messy, which is normal for biological field experiments because you can't control everything in soil the way you can in a lab.

```{r}
clay$per.clay <- as.factor(clay$per.clay)
clay$days <- as.factor(clay$days)

model_clay <- aov(ralstonia ~ per.clay + days, data = clay)
summary.aov(model_clay)
cv.model(model_clay)
```

```
            Df Sum Sq Mean Sq F value   Pr(>F)
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days         5  70.88  14.176    5.99 0.000184 ***
Residuals   53 125.44   2.367
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
[1] 26.48258
```

Regression Model

```{r}
lm_clay <- lm(ralstonia ~ per.clay + days, data = clay)
summary.lm(lm_clay)
```

```
Call:
lm(formula = ralstonia ~ per.clay + days, data = clay)

Residuals:
    Min      1Q  Median      3Q     Max 
-4.3900 -0.6654  0.2163  0.6176  2.8121 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)   7.9211     0.7454  10.627 9.62e-15 ***
per.clay6    -0.1447     0.8882  -0.163 0.871236    
per.clay8     0.5887     0.8882   0.663 0.510355    
per.clay10    0.6658     0.8882   0.750 0.456785    
per.clay12    0.2532     0.8882   0.285 0.776730    
per.clay14   -0.6677     0.7692  -0.868 0.389310    
per.clay20    0.5360     0.8882   0.603 0.548773    
per.clay22   -1.9850     1.2837  -1.546 0.127980    
per.clay24   -0.8595     0.7964  -1.079 0.285336    
per.clay28   -2.4869     1.1048  -2.251 0.028562 *  
per.clay32   -3.3682     0.8882  -3.792 0.000385 ***
days15       -0.4902     0.6034  -0.812 0.420253    
days29       -2.2136     0.6220  -3.559 0.000795 ***
days43       -2.4883     0.6409  -3.882 0.000288 ***
days58       -2.3386     0.6589  -3.549 0.000819 ***
days73       -2.5180     0.6589  -3.821 0.000350 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.538 on 53 degrees of freedom
Multiple R-squared:  0.5487,	Adjusted R-squared:  0.421 
F-statistic: 4.296 on 15 and 53 DF,  p-value: 3.849e-05
```

The baseline reference level is 4% clay on day 2, represented by the intercept of 7.9211. This means the expected Ralstonia level at 4% clay on day 2 is 7.92 units.

ralstonia=β0+βclay+βdays+ϵ

ralstonia = 7.9211 - 0.1447(per.clay6) + 0.5887(per.clay8) + 0.6658(per.clay10) + 0.2532(per.clay12) - 0.6677(per.clay14) + 0.5360(per.clay20) - 1.9850(per.clay22) - 0.8595(per.clay24) - 2.4869(per.clay28) - 3.3682(per.clay32) - 0.4902(days15) - 2.2136(days29) - 2.4883(days43) - 2.3386(days58) - 2.5180(days73) + ε

CLAY TREATMENT FACTORS:

per.clay6: estimate = -0.1447, p = 0.871236— NOT significant. 6% clay does not meaningfully change Ralstonia levels compared to 4% clay.
per.clay8: estimate = +0.5887, p = 0.510355 — NOT significant. 8% clay does not meaningfully change Ralstonia levels compared to 4% clay.
per.clay10: estimate = +0.6658, p = 0.456785 — NOT significant. 10% clay does not meaningfully change Ralstonia levels compared to 4% clay.
per.clay12: estimate = +0.2532, p = 0.776730— NOT significant. 12% clay does not meaningfully change Ralstonia levels compared to 4% clay.
per.clay14: estimate = -0.6677, p = 0.389310— NOT significant. 14% clay does not meaningfully change Ralstonia levels compared to 4% clay.
per.clay20: estimate = +0.5360, p = 0.548773— NOT significant. 20% clay does not meaningfully change Ralstonia levels compared to 4% clay.
per.clay22: estimate = -1.9850, p = 0.127980 — NOT significant. 22% clay does not meaningfully change Ralstonia levels compared to 4% clay, although the estimate is getting larger suggesting a trend toward lower bacteria levels at higher clay percentages.
per.clay24: estimate = -0.8595, p = 0.285336— NOT significant. 24% clay does not meaningfully change Ralstonia levels compared to 4% clay.
per.clay28: estimate = -2.4869, p = 0.028562 — SIGNIFICANT. 28% clay significantly reduces Ralstonia levels by approximately 2.49 units compared to 4% clay.
per.clay32: estimate = -3.3682, p = 0.000385 — HIGHLY SIGNIFICANT. 32% clay reduces Ralstonia levels by approximately 3.37 units compared to 4% clay. This is the strongest and most significant clay treatment effect in the entire model.

BLOCKING FACTOR — DAYS:

days15: estimate = -0.4902, p = 0.420253— NOT significant. Day 15 does not differ significantly from the baseline day 2. Bacteria levels had not yet changed meaningfully by day 15.
days29: estimate = -2.2136, p = 0.0008 — HIGHLY SIGNIFICANT. By day 29 Ralstonia levels had dropped by approximately 2.2136 units compared to day 2. This marks the point where bacteria
levels began declining significantly.
days43: estimate = -2.4883, p = 0.0003 — HIGHLY SIGNIFICANT. Day 43 shows Ralstonia levels approximately 2.4883 units lower than day 2, continuing the downward trend.
days58: estimate = -2.3386, p = 0.0008 — HIGHLY SIGNIFICANT. Day 58 shows Ralstonia levels approximately 2.3386 units lower than day 2, remaining consistently low.
days73: estimate = -2.5180, p = 0.0004 — HIGHLY SIGNIFICANT. Day 73 shows the largest time effect, with Ralstonia levels approximately 2.5180 units lower than day 2. Bacteria levels
remained low through the end of the experiment.

MODEL FIT:
Multiple R squared = 0.5487
Adjusted R squared = 0.4210
Overall p-value = 3.849e-05
The model explains approximately 54.9% of the variation in Ralstonia levels and is highly significant overall. The gap between Multiple R squared and Adjusted R squared suggests that several clay treatment levels add limited explanatory power since many are not significantly different from the baseline. The remaining 45.1% of variation is attributed to natural biological variability that the model could not capture.

Assumptions

```{r}
shapiro.test(residuals(model_clay))   # normality of residuals
bartlett.test(ralstonia ~ per.clay, data = clay)  # equal variances

```

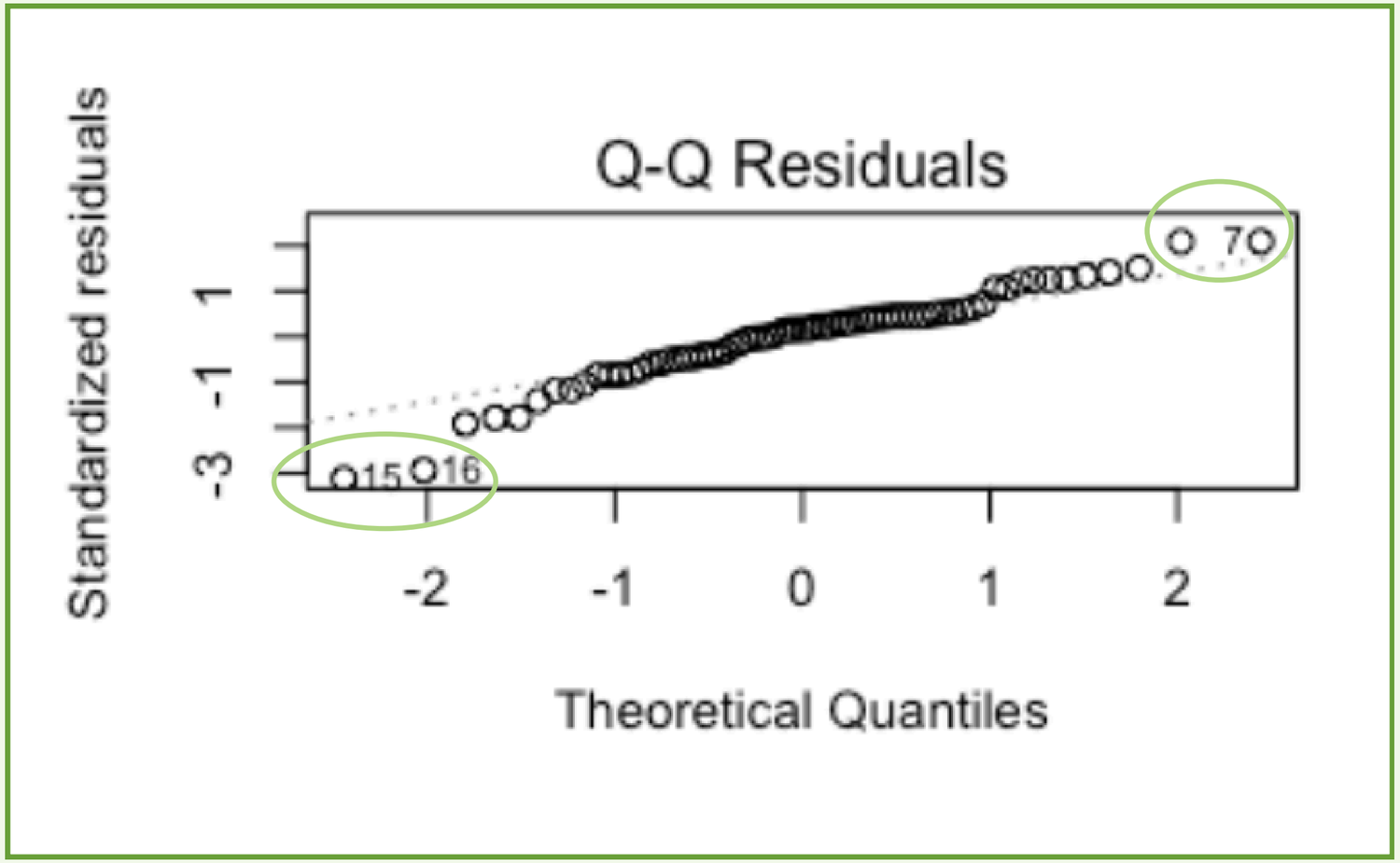
```
        Shapiro-Wilk normality test

data:  residuals(model_clay)
W = 0.94906, p-value = 0.006994


        Bartlett test of homogeneity of variances

data:  ralstonia by per.clay
Bartlett's K-squared = 49.229, df = 10, p-value = 3.699e-07
```

T**he Shapiro-Wilk test returned p = 0.006994**, which is less than 0.05. This means we reject H0 —> the residuals are not perfectly normal, indicating a violation of the normality assumption.---> *we will look at the QQ-plot (slide 7).*

Bartlett's test returned p = 3.699e-07, which is well below 0.05. This means we reject H0 —> variances are not equal across clay treatments, indicating a violation of the homogeneity of variance assumption.---> *we will look at the Scale Location plot (slide 8).*

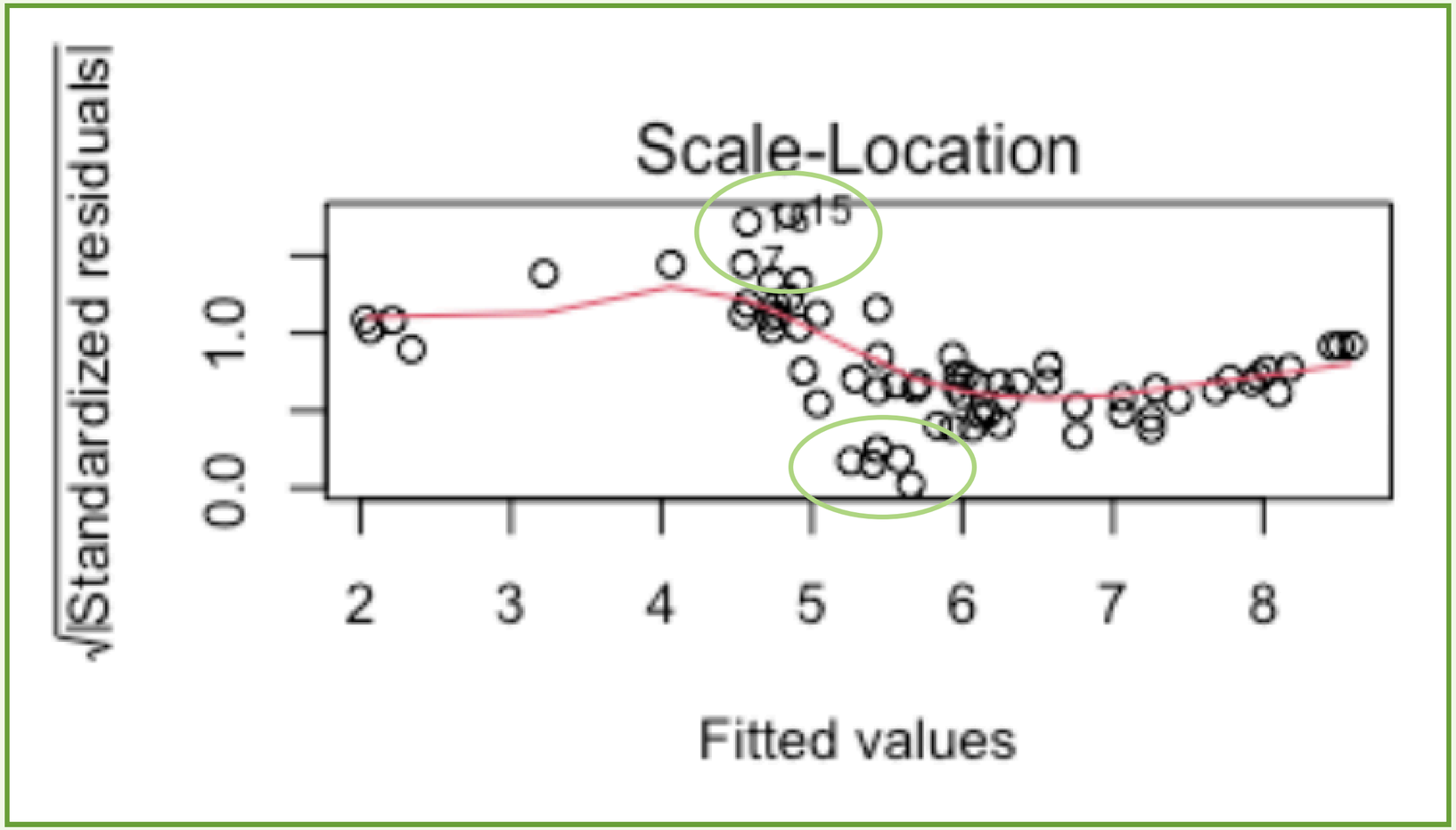
ANOVA can generally handle mild violations, it is expected in biological data for natural variability. These violations are common and expected in biological field experiments where bacterial levels naturally vary between soil samples in ways we cannot fully control. Our results should therefore be interpreted with appropriate caution.

Q-Q plot



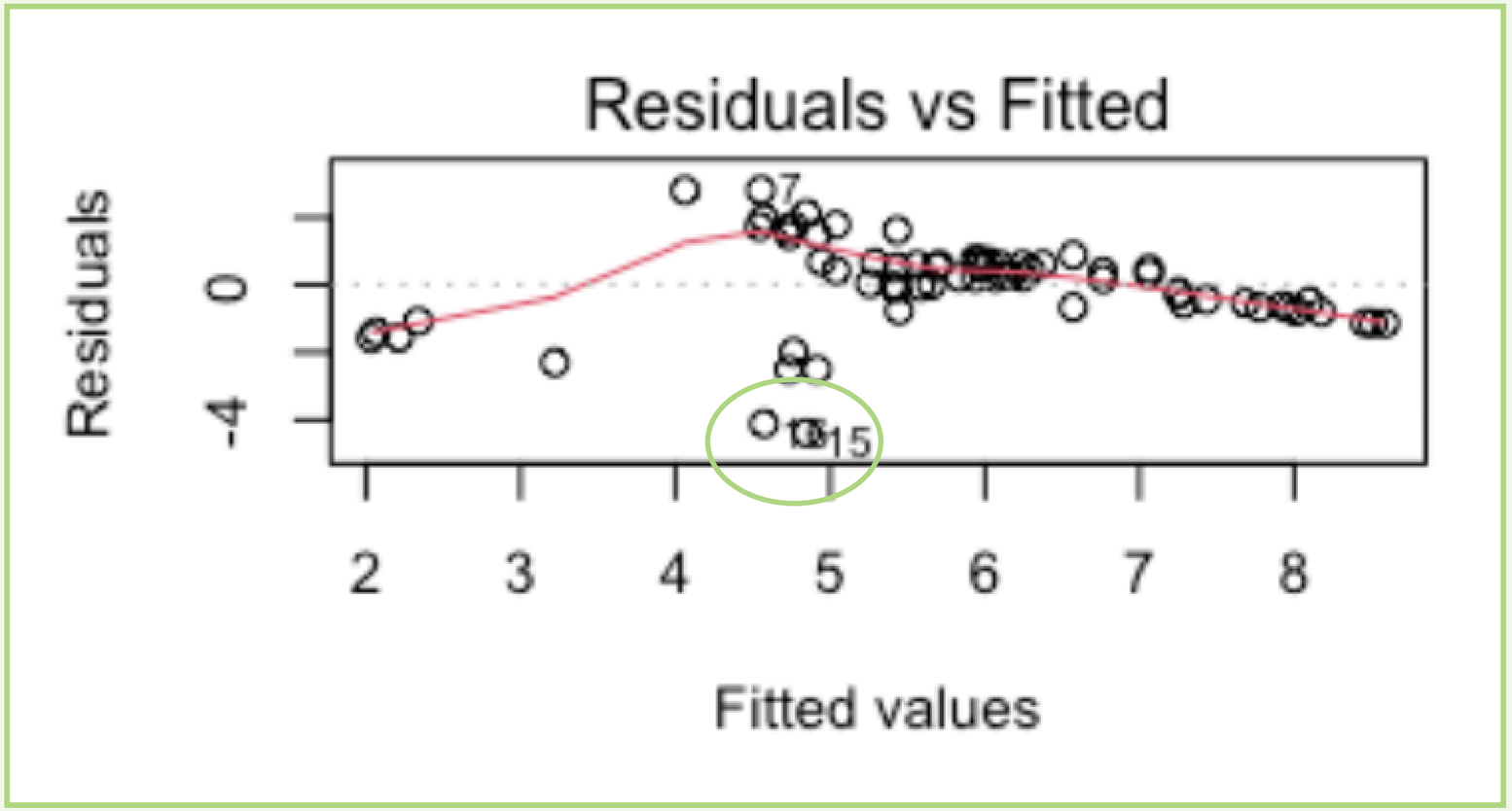
Most of our points do follow the line through the middle, which is encouraging. But points 15 and 16 deviate sharply at the lower left tail, and point 70 deviates at the upper right. These three observations are the main source of the normality violation, it is being driven by a small number of extreme observations rather than the entire dataset

Scale-Location



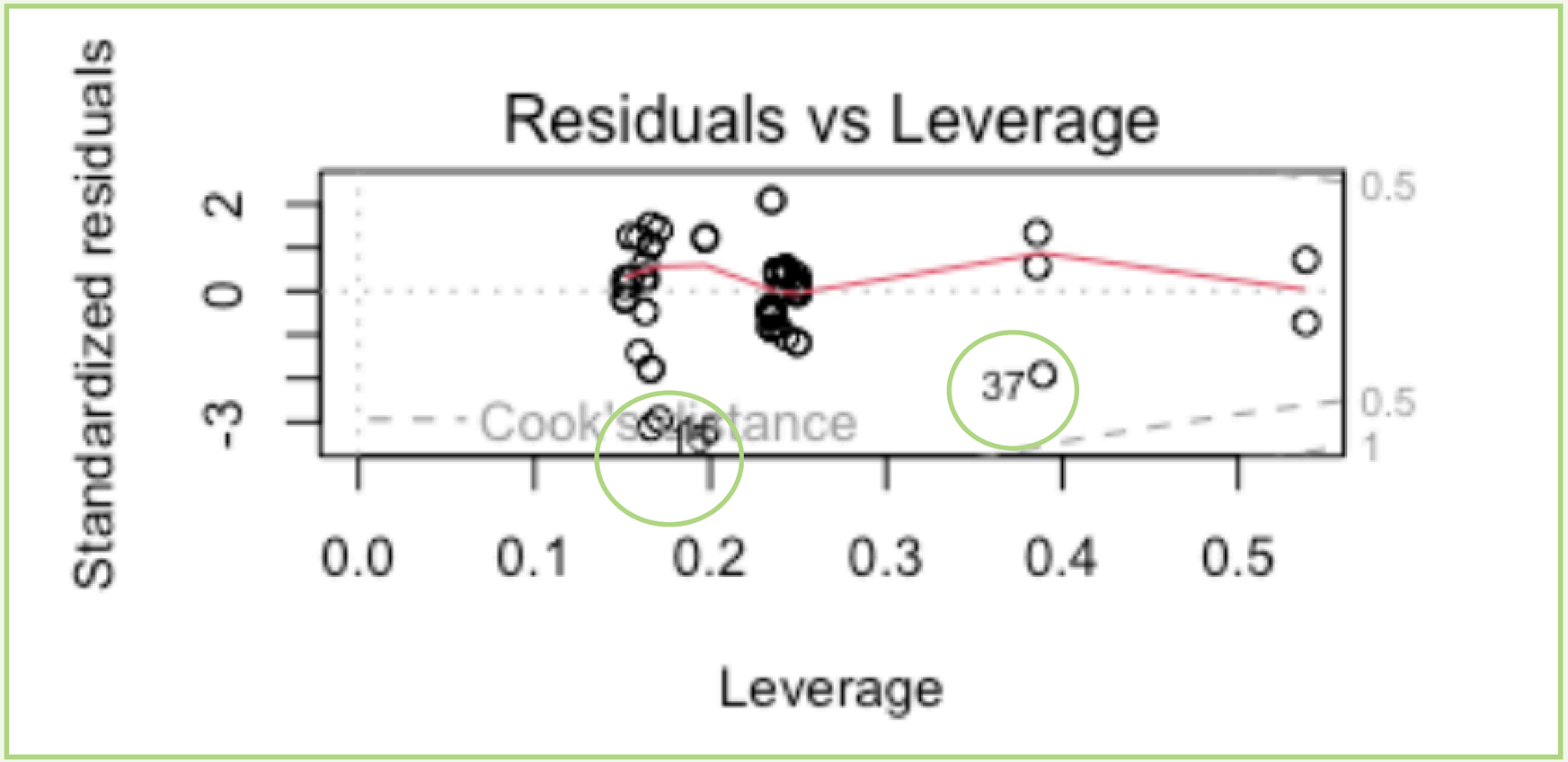
The red line should be roughly horizontal if variances are equal across treatments. Our line curves starting high, dipping in the middle, and rising again on the right. This confirms the Bartlett test result. Points 16 and 15 are flagged again.

Residuals vs Fitted



The Residuals vs Fitted plot checks overall model fit. The red line should be flat and the points randomly scattered. Our line has a slight curve and points 13 and 15 appear as large negative outliers pulling the line down around fitted values of 4 to 5. This tells us the model struggles to fit some observations accurately

Residuals vs Leverage



The Residuals vs Leverage plot checks for influential observations ,points that could be distorting the entire model. Point 37 is close to the Cook's distance dashed line, and there is another observation on the far right with high leverage around 0.5. These points have some influence on the model. However since no points have clearly crossed outside the Cook's distance boundary they are not considered severely influential enough to invalidate our results

Pairwise Comparison: TukeyHSD

The 32% clay treatment had significantly lower Ralstonia levels compared to 7 other treatments:

All other pairwise comparisons were not significant,meaning those clay percentages produced similar Ralstonia levels to each other.

Overall, 32% clay stands out as the only treatment that consistently and significantly reduces Ralstonia levels, making it the most effective clay percentage tested.

```{r}
TukeyHSD(model_clay, "per.clay")
```

```
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = ralstonia ~ per.clay + days, data = clay)

$per.clay
             diff       lwr        upr     p adj
6-4   -0.14466667 -3.131579  2.8422459 1.0000000
8-4    0.58866667 -2.398246  3.5755792 0.9998465
10-4   0.66583333 -2.321079  3.6527459 0.9995407
12-4   0.25316667 -2.733746  3.2400792 1.0000000
14-4  -0.66766667 -3.254409  1.9190755 0.9983845
20-4   0.53600000 -2.450913  3.5229126 0.9999346
22-4  -0.55533333 -4.779466  3.6687990 0.9999965
24-4  -0.70883333 -3.380409  1.9627425 0.9979671
28-4  -1.71333333 -5.371539  1.9448725 0.8855873
32-4  -3.36816667 -6.355079 -0.3812541 0.0153500
8-6    0.73333333 -2.253579  3.7202459 0.9989415
10-6   0.81050000 -2.176413  3.7974126 0.9975527
12-6   0.39783333 -2.589079  3.3847459 0.9999960
14-6  -0.52300000 -3.109742  2.0637422 0.9998068
20-6   0.68066667 -2.306246  3.6675792 0.9994432
22-6  -0.41066667 -4.634799  3.8134656 0.9999998
24-6  -0.56416667 -3.235742  2.1074092 0.9997152
28-6  -1.56866667 -5.226873  2.0895392 0.9315805
32-6  -3.22350000 -6.210413 -0.2365874 0.0243991
10-8   0.07716667 -2.909746  3.0640792 1.0000000
12-8  -0.33550000 -3.322413  2.6514126 0.9999992
14-8  -1.25633333 -3.843076  1.3304088 0.8607042
20-8  -0.05266667 -3.039579  2.9342459 1.0000000
22-8  -1.14400000 -5.368132  3.0801323 0.9975914
24-8  -1.29750000 -3.969076  1.3740758 0.8607243
28-8  -2.30200000 -5.960206  1.3562059 0.5715454
32-8  -3.95683333 -6.943746 -0.9699208 0.0019927
12-10 -0.41266667 -3.399579  2.5742459 0.9999943
14-10 -1.33350000 -3.920242  1.2532422 0.8114240
20-10 -0.12983333 -3.116746  2.8570792 1.0000000
22-10 -1.22116667 -5.445299  3.0029656 0.9959075
24-10 -1.37466667 -4.046242  1.2969092 0.8131306
28-10 -2.37916667 -6.037373  1.2790392 0.5236164
32-10 -4.03400000 -7.020913 -1.0470874 0.0015016
14-12 -0.92083333 -3.507576  1.6659088 0.9801214
20-12  0.28283333 -2.704079  3.2697459 0.9999999
22-12 -0.80850000 -5.032632  3.4156323 0.9998822
24-12 -0.96200000 -3.633576  1.7095758 0.9784341
28-12 -1.96650000 -5.624706  1.6917059 0.7702707
32-12 -3.62133333 -6.608246 -0.6344208 0.0065599
20-14  1.20366667 -1.383076  3.7904088 0.8896465
22-14  0.11233333 -3.838981  4.0636473 1.0000000
24-14 -0.04116667 -2.256320  2.1739870 1.0000000
28-14 -1.04566667 -4.385136  2.2938031 0.9923341
32-14 -2.70050000 -5.287242 -0.1137578 0.0337916
22-20 -1.09133333 -5.315466  3.1327990 0.9983715
24-20 -1.24483333 -3.916409  1.4267425 0.8888060
28-20 -2.24933333 -5.907539  1.4088725 0.6042942
32-20 -3.90416667 -6.891079 -0.9172541 0.0024129
24-22 -0.15350000 -4.160864  3.8538637 1.0000000
28-22 -1.15800000 -5.880723  3.5647235 0.9989532
32-22 -2.81283333 -7.036966  1.4112990 0.4887204
28-24 -1.00450000 -4.410104  2.4011043 0.9951971
32-24 -2.65933333 -5.330909  0.0122425 0.0520320
32-28 -1.65483333 -5.313039  2.0033725 0.9059966
```

Concluding Dataset 1

1

32% clay was the only treatment that significantly reduced Ralstonia levels, making it the most effective clay percentage tested.

2

Both ANOVA assumption were violated,normality and equal variance, but results remain valid given the biological nature of the data.

3

The blocking factor days was **highly significant**, confirming that time variation was correctly controlled for in the experimental design.

DATASET 2: MELON

What is this experiment about?
A scientist wanted to know which melon variety produces the best yield.Using a Latin-Square Design, they tested 4 different melon varieties (V1, V2, V3, V4) in a field laid out as a 4 x 4 grid with 4 rows and 4 columns, giving 16 total observations. The dataset is perfectly balanced.

Why were row and col chosen as blocking factors and not treatment factors?
Because the researcher was not interested in studying the effect of field position itself — they wanted to study melon variety. Row and column were included only to control for natural variation across the field. If we ignored them, that spatial variation would end up in the error term and make it harder to detect real variety differences. Two blocking factors are used instead of one because variation can run in both the horizontal and vertical direction across a physical field — this is what makes it a Latin Square Design rather than a standard RCBD.

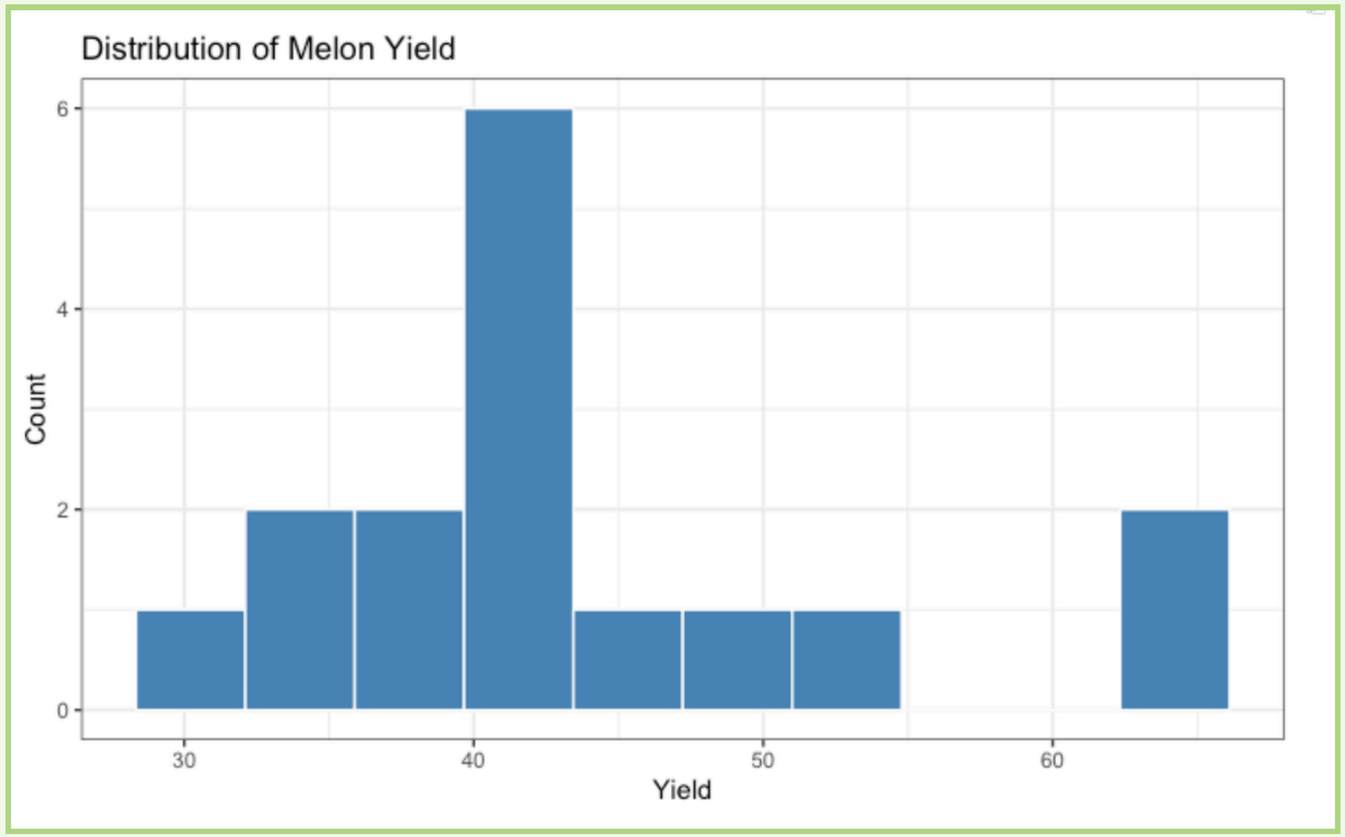
Summary:
The melon dataset contains 16 observations of melon yield under different variety treatments and field positions. The variable variety represents the treatment factor, while row and col represent two blocking factors controlling for field variation in both directions. The response variable yield measures melon production.

```
'data.frame':   16 obs. of  4 variables:
 $ row    : int  1 1 1 1 2 2 2 2 3 3 ...
 $ col    : int  1 2 3 4 1 2 3 4 1 2 ...
 $ variety: Factor w/ 4 levels "V1","V2","V3",..: 1 4 2 3 2 3 4 1 3 2 ...
 $ yield  : int  45 29 37 38 50 53 41 40 43 41 ...
      row            col       variety     yield
 Min.   :1.00   Min.   :1.00   V1:4    Min.   :29.00
 1st Qu.:1.75   1st Qu.:1.75   V2:4    1st Qu.:37.75
 Median :2.50   Median :2.50   V3:4    Median :41.00
 Mean   :2.50   Mean   :2.50   V4:4    Mean   :43.44
 3rd Qu.:3.25   3rd Qu.:3.25           3rd Qu.:46.25
 Max.   :4.00   Max.   :4.00           Max.   :63.00
```

What are the variables?
variety — this is what is being tested. Think of it as the different types of melon being compared. 4 different varieties were used.
row — the field was divided into 4 horizontal rows. This was included because different parts of the field may have better soil or more sunlight, and they didn't want that to affect the variety comparison.
col — the field was also divided into 4 vertical columns for the same reason. Including both row and col controls for variation in two directions simultaneously.
yield — this is what they measured. How much melon was produced. Higher number = better yield.

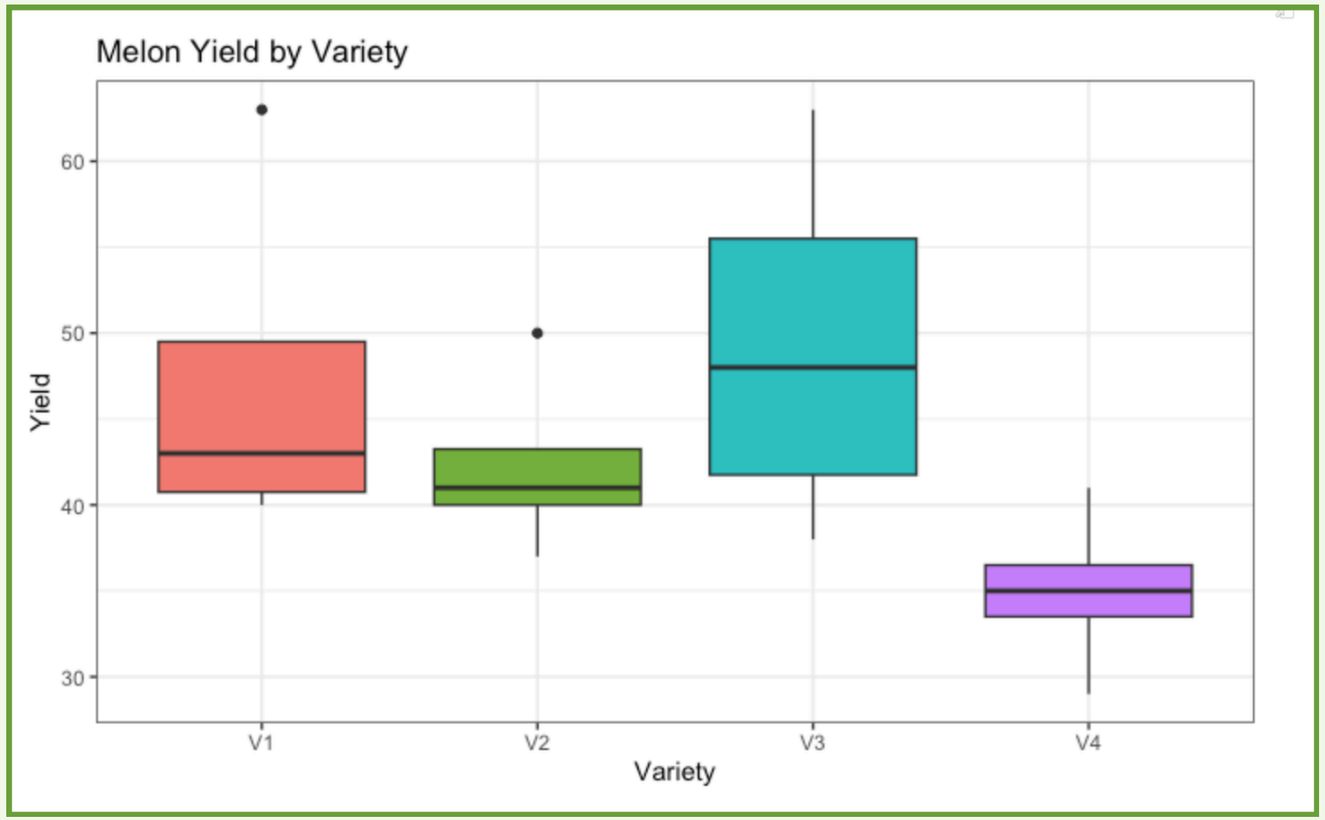
The response variable yield ranges from a minimum of 29 to a maximum of 63, with an overall mean of 43.44. The median of 41 is slightly below the mean, suggesting one or two higher yielding observations may be pulling the average up slightly.
Row and column are both integers ranging from 1 to 4, representing the grid position of each plot in the field. Variety is a factor with 4 levels (V1, V2, V3, and V4) each appearing exactly 4 times confirming the balanced Latin Square structure.

Looking at the Data: Plot 1 — Histogram



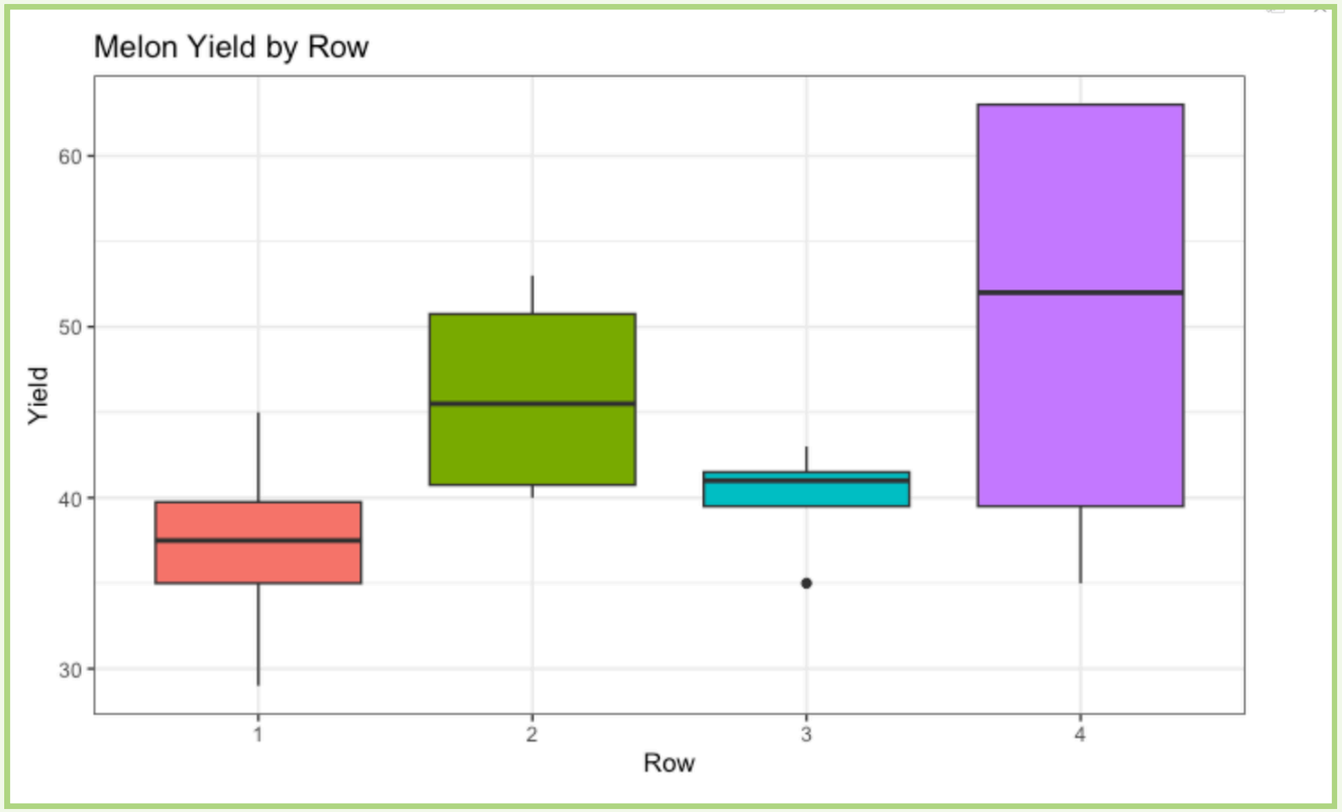
Shows the overall distribution of melon yield across the entire experiment. Most yields cluster between 38 and 45 with the distribution centered around 41. Two observations near 63 appear on the far right, pulling the mean slightly above the median. This tells us the data is slightly right skewed with a small number of high yielding plots.

Looking at the Data: Plot 2 — Boxplot by variety



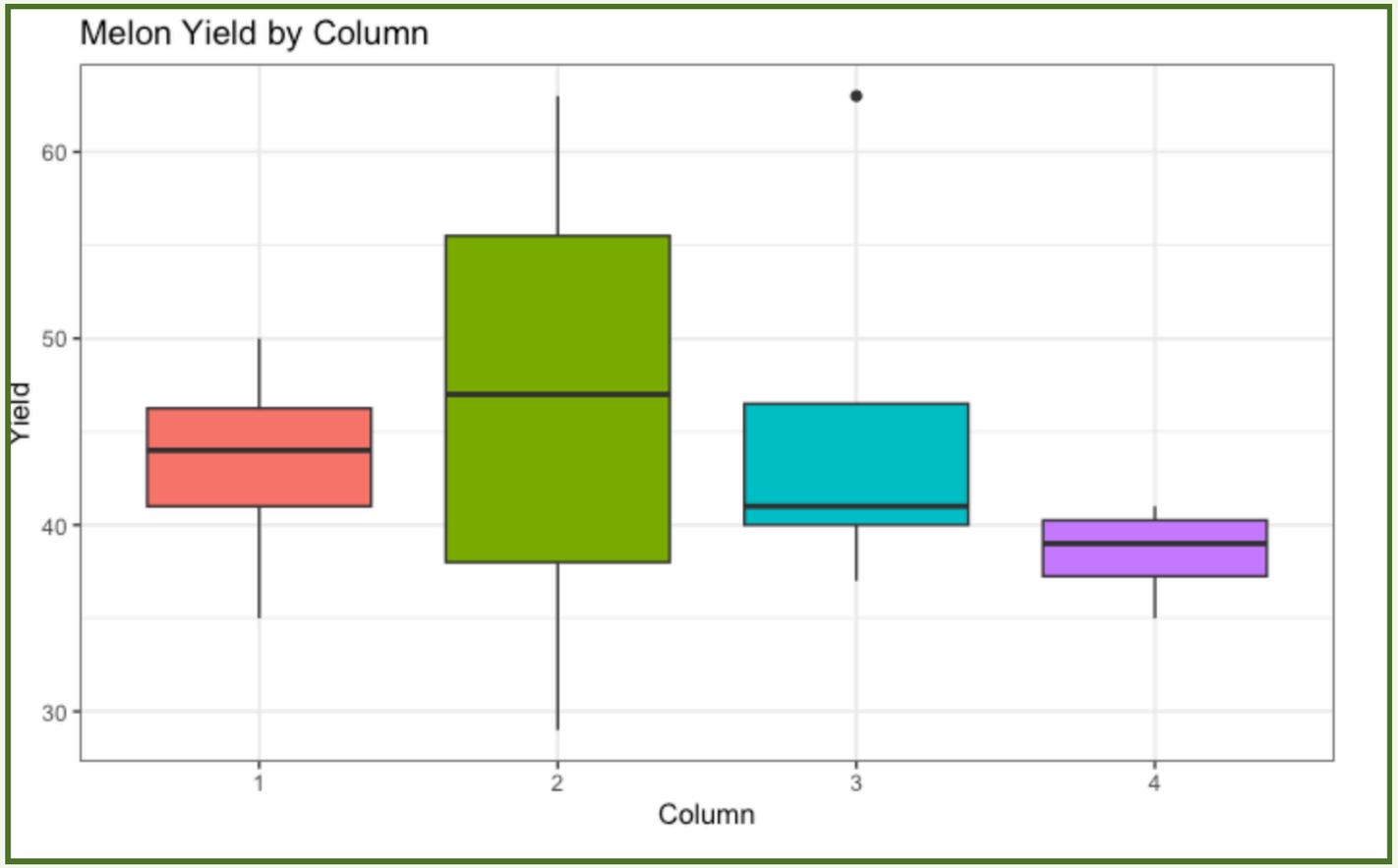
Shows how yield varies across the 4 melon varieties. V3 has the highest median yield and widest spread, while V4 has the lowest median and smallest spread. V1 has one outlier above 60 pulling its range upward. The boxes overlap considerably across all varieties, giving us an early visual hint that differences between varieties may not be statistically significant.

Looking at the Data: Plot 3 — Boxplot by row



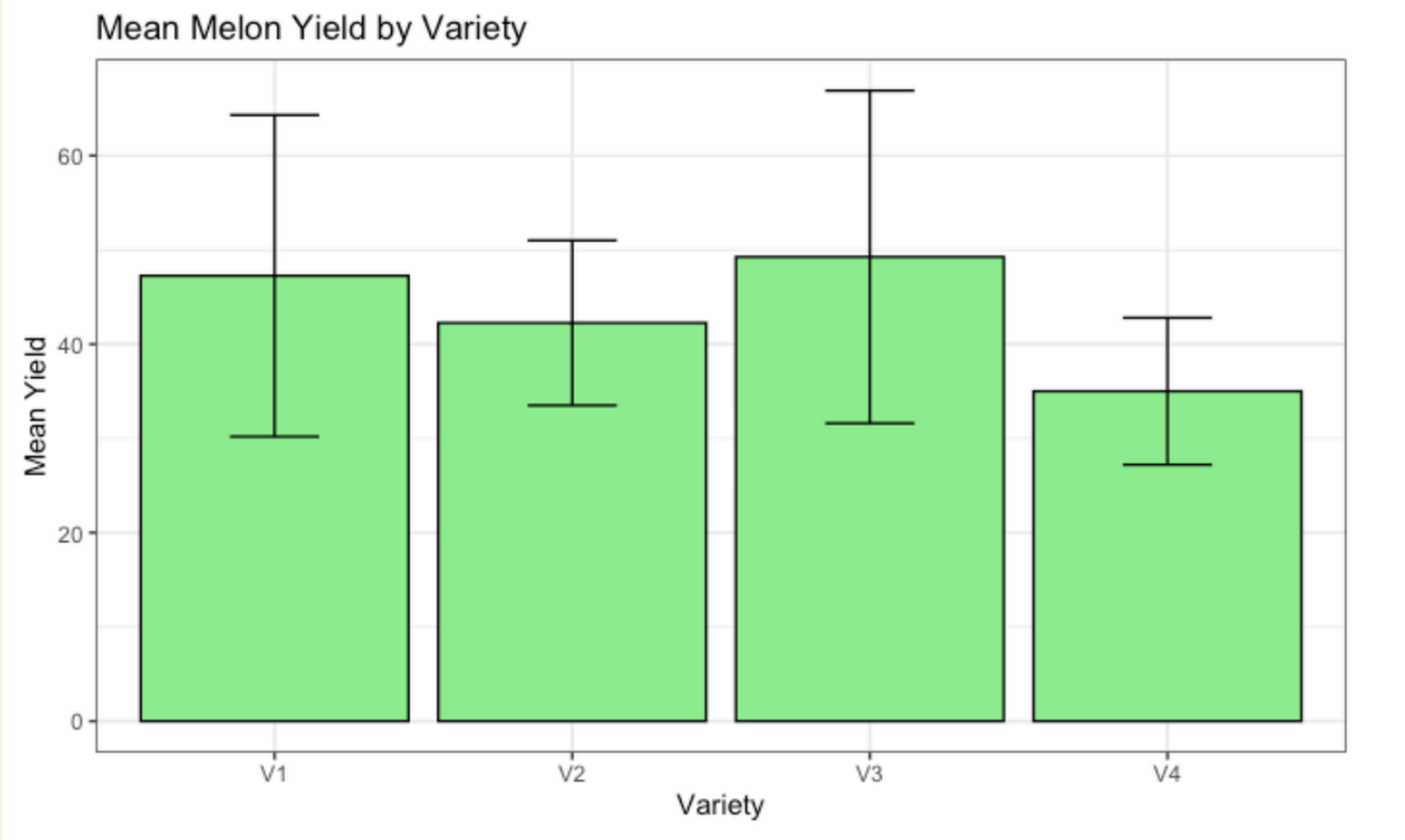
Shows how yield varies across the 4 field rows. Row 4 visually stands out with the highest median yield and widest spread, suggesting that part of the field was more productive regardless of which variety was planted there. Row 1 has the lowest median and tightest spread along with 3. Row 3 has one low outlier near 35. This gives us an early indication that row position may influence yield, justifying its inclusion as a blocking factor.

Looking at the Data: Plot 4 — Boxplot by column



Shows how yield varies across the 4 field columns. Column 2 has the widest spread and highest median while column 4 has the lowest. However the ranges overlap considerably and column 3 has one outlier above 60. The variation between columns appears smaller than between rows, suggesting column position has less influence on yield than row position.

Looking at the Data: Plot 5 — Bar chart of means



Shows the average yield for each variety with error bars representing variability within each group. V3 has the highest mean yield while V4 has the lowest. However the error bars are very wide and overlap heavily across all four varieties , this visually tells us before running any test that no variety clearly outperforms the others.

ANOVA: Latin Square

```
## ANOVA
```{r}
melon$row <- as.factor(melon$row)
melon$col <- as.factor(melon$col)
melon$variety <- as.factor(melon$variety)

model_melon <- aov(yield ~ variety + row + col, data = melon)
summary.aov(model_melon)
cv.model(model_melon)
```
```

```
            Df Sum Sq Mean Sq F value Pr(>F)
variety      3  483.7  161.23   3.248  0.102
row          3  426.2  142.06   2.862  0.126
col          3  152.2   50.73   1.022  0.447
Residuals    6  297.9   49.65
[1] 16.22096
```

$Y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_k + \varepsilon_{ijk}$

$variety_{ijk} = \mu + \alpha_i + \beta_j + \gamma_k + \varepsilon_{ijk}$

Y_{ijk} = the observed yield for the ith variety in the jth row and kth column
μ = the overall grand mean of all yield measurements
α_i = the effect of the ith variety (i = V1, V2, V3, V4)
β_j = the effect of the jth row block (j = 1, 2, 3, 4)
γ_k = the effect of the kth column block (k = 1, 2, 3, 4)
ε_{ijk} = the random error term for each observation

yield = grand mean + effect of variety + effect of row + effect of column + random error (ε)

The key difference from the clay equation is the extra term γ_k —> this represents the second blocking factor, column. Having two blocking factors is what makes this a Latin Square Design rather than a standard RCBD.

T**he residual mean square is 49.65** — this is the size of our random error term epsilon. Compare this to clay's residual mean square of 2.367, melon's is much larger, which is partly why it was harder to detect significant differences.

ANOVA: Latin Square

Does melon variety actually affect yield, or are any differences just random chance?

variety (F = 3.248, df = 3, p = 0.102): Since p is greater than 0.05 we fail to reject H0. Variety did not have a statistically significant effect on melon yield. We cannot conclude that any variety outperforms the others based on this experiment.

Row (F = 2.862, df = 3, p = 0.126): Since p is greater than 0.05 we fail to reject H0. Row position did not significantly affect yield, meaning horizontal variation across the field was not strong enough to reach significance.

Column (F = 1.022, df = 3, p = 0.447): Since p is greater than 0.05 we fail to reject H0. Column position did not significantly affect yield either, meaning vertical variation across the field was minimal.

Residuals: The residual mean square is 49.65, which represents the unexplained variation in the model after accounting for variety, row, and column effects.

```
## ANOVA
```{r}
melon$row <- as.factor(melon$row)
melon$col <- as.factor(melon$col)
melon$variety <- as.factor(melon$variety)

model_melon <- aov(yield ~ variety + row + col, data = melon)
summary.aov(model_melon)
cv.model(model_melon)
```
```

```
            Df Sum Sq Mean Sq F value Pr(>F)
variety      3  483.7  161.23   3.248  0.102
row          3  426.2  142.06   2.862  0.126
col          3  152.2   50.73   1.022  0.447
Residuals    6  297.9   49.65
[1] 16.22096
```

CV% = 16.22%:The coefficient of variation is 16.22%, indicating moderate experimental variability. This is acceptable for field crop data and is actually better than the clay dataset which had a CV of 26.48%, suggesting the melon experiment was more controlled.

Conclusion: None of the factors reached statistical significance. This does not mean the experiment failed, it means we do not have sufficient evidence to conclude differences exist based on this sample size.

Regression Coefficients

The baseline is Variety V1 in Row 1 and Column 1, with an expected yield of 40.875.

```
# Regression coefficient
```{r}
lm_melon <- lm(yield ~ variety + row + col, data = melon)
summary.lm(lm_melon)
```

Call:
lm(formula = yield ~ variety + row + col, data = melon)

Residuals:
   Min   1Q Median   3Q   Max
-6.875 -3.000 -1.000 4.250 8.375

Coefficients:

	Estimate	Std. Error	t value	Pr(>|t|)	
(Intercept)	40.875	5.570	7.338	0.000328	***
varietyV2	-5.000	4.982	-1.004	0.354331	
varietyV3	2.000	4.982	0.401	0.702014	
varietyV4	-12.250	4.982	-2.459	0.049205	*
row2	8.750	4.982	1.756	0.129569	
row3	2.750	4.982	0.552	0.600915	
row4	13.250	4.982	2.659	0.037552	*
col2	3.250	4.982	0.652	0.538371	
col3	2.250	4.982	0.452	0.667420	
col4	-4.750	4.982	-0.953	0.377209	

Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 7.046 on 6 degrees of freedom
Multiple R-squared: 0.781,	Adjusted R-squared: 0.4524
F-statistic: 2.377 on 9 and 6 DF, p-value: 0.1521

yield = 40.875 - 5.000(varietyV2) + 2.000(varietyV3) - 12.250(varietyV4) + 8.750(row2)+ 2.750(row3) + 13.250(row4) + 3.250(col2) + 2.250(col3) - 4.750(col4) + ε

VARIETY EFFECTS:

varietyV2: estimate = -5.000, p = 0.354 — NOT significant.V2 produces approximately 5 units less yield than V1 but this difference is not statistically meaningful.
varietyV3: estimate = +2.000, p = 0.702 — NOT significant.V3 produces approximately 2 units more yield than V1 but this difference is not statistically meaningful.
varietyV4: estimate = -12.250, p = 0.049 — SIGNIFICANT. V4 is the only variety that significantly differs from the baseline. It produces approximately 12.25 units less yield than V1. This is the strongest variety effect in the model.

ROW EFFECTS:

row2: estimate = +8.750, p = 0.130 — NOT significant. Row 2 produces approximately 8.75 units more yield than row 1 but this is not statistically significant.
row3: estimate = +2.750, p = 0.601 — NOT significant. Row 3 does not meaningfully differ from row 1.
row4: estimate = +13.250, p = 0.038 — SIGNIFICANT. Row 4 is the only row that significantly differs from the baseline. Plots in row 4 produced approximately 13.25 units more yield than row 1, suggesting that part of the field was more productive regardless of variety planted there.

COLUMN EFFECTS:

col2: estimate = +3.250, p = 0.538 — NOT significant.
col3: estimate = +2.250, p = 0.667 — NOT significant.
col4: estimate = -4.750, p = 0.377 — NOT significant.
None of the column positions significantly affected yield, confirming that vertical variation across the field was minimal.

MODEL FIT:

Multiple R squared = 0.781
Adjusted R squared = 0.452
Overall p-value = 0.152

The model appears to explain 78.1% of variation on the surface but the Adjusted R squared of 0.452 tells a more honest story.
The large gap between the two values tells us the model is overfitting, we have 9 predictors but only 16 observations,
which is too many predictors for such a small dataset. The overall model is not significant (p = 0.152), consistent with the ANOVA results.
```

Assumptions

Shapiro-Wilk (W = 0.950, p = 0.4918):
Since p is greater than 0.05 we fail to reject H0. The residuals are approximately normally distributed and the normality assumption is met. This is a good result and contrasts with the clay dataset where normality was violated. (*Refer to QQ Plot slide 26)*

Bartlett's test (K squared = 2.686, df = 3, p = 0.4426):
Since p is greater than 0.05 we fail to reject H0. Variances are equal across melon varieties and the homogeneity of variance assumption is met. (Refer to

Both assumptions are fully satisfied for the melon dataset. This means our ANOVA results are statistically reliable and no violations need to be noted — unlike the clay dataset where both assumptions were violated.

```
# Assumption check
```{r}
shapiro.test(residuals(model_melon))
bartlett.test(yield ~ variety, data = melon)
```
```

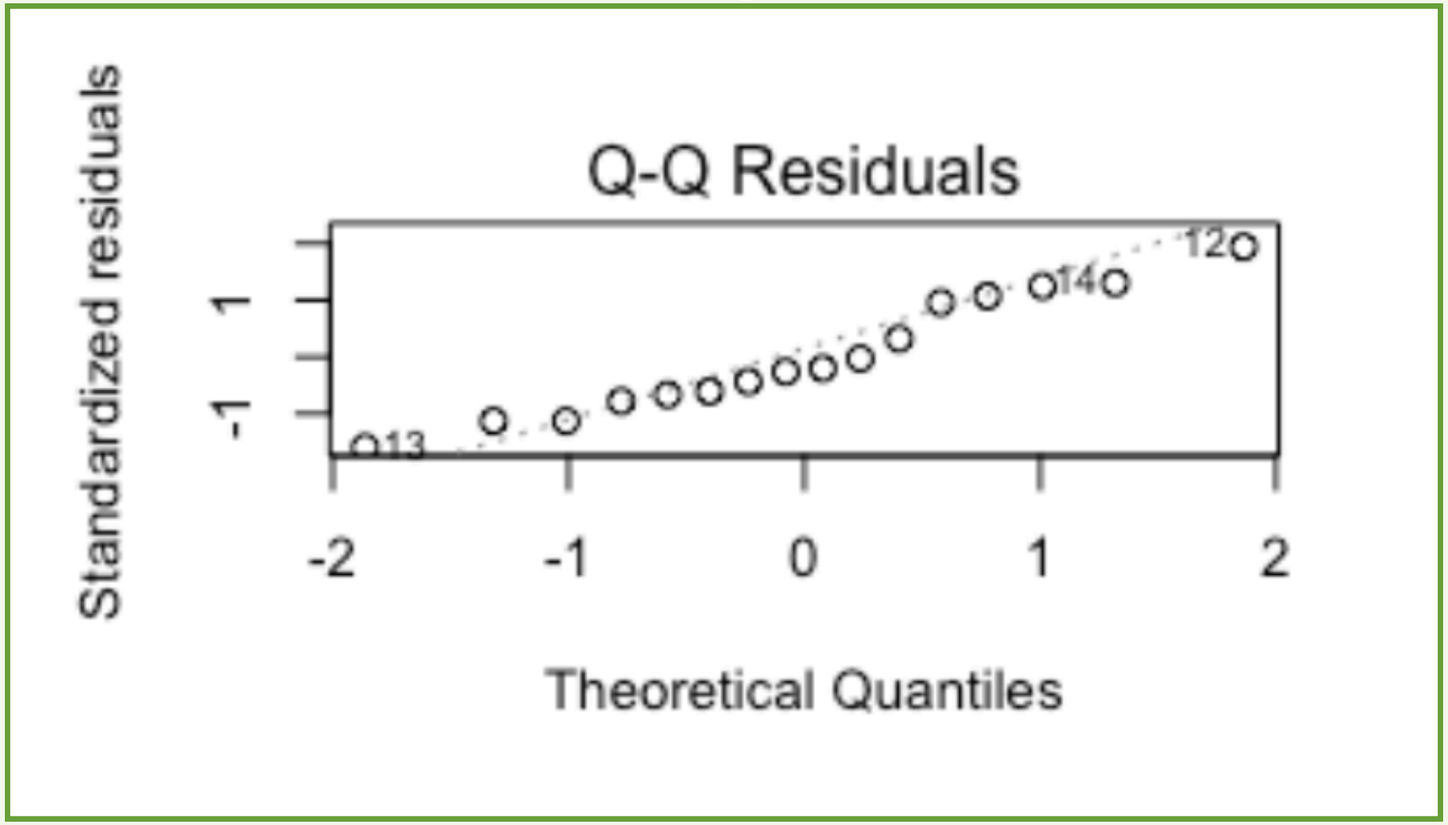
```
        Shapiro-Wilk normality test

data:  residuals(model_melon)
W = 0.95013, p-value = 0.4918


        Bartlett test of homogeneity of variances

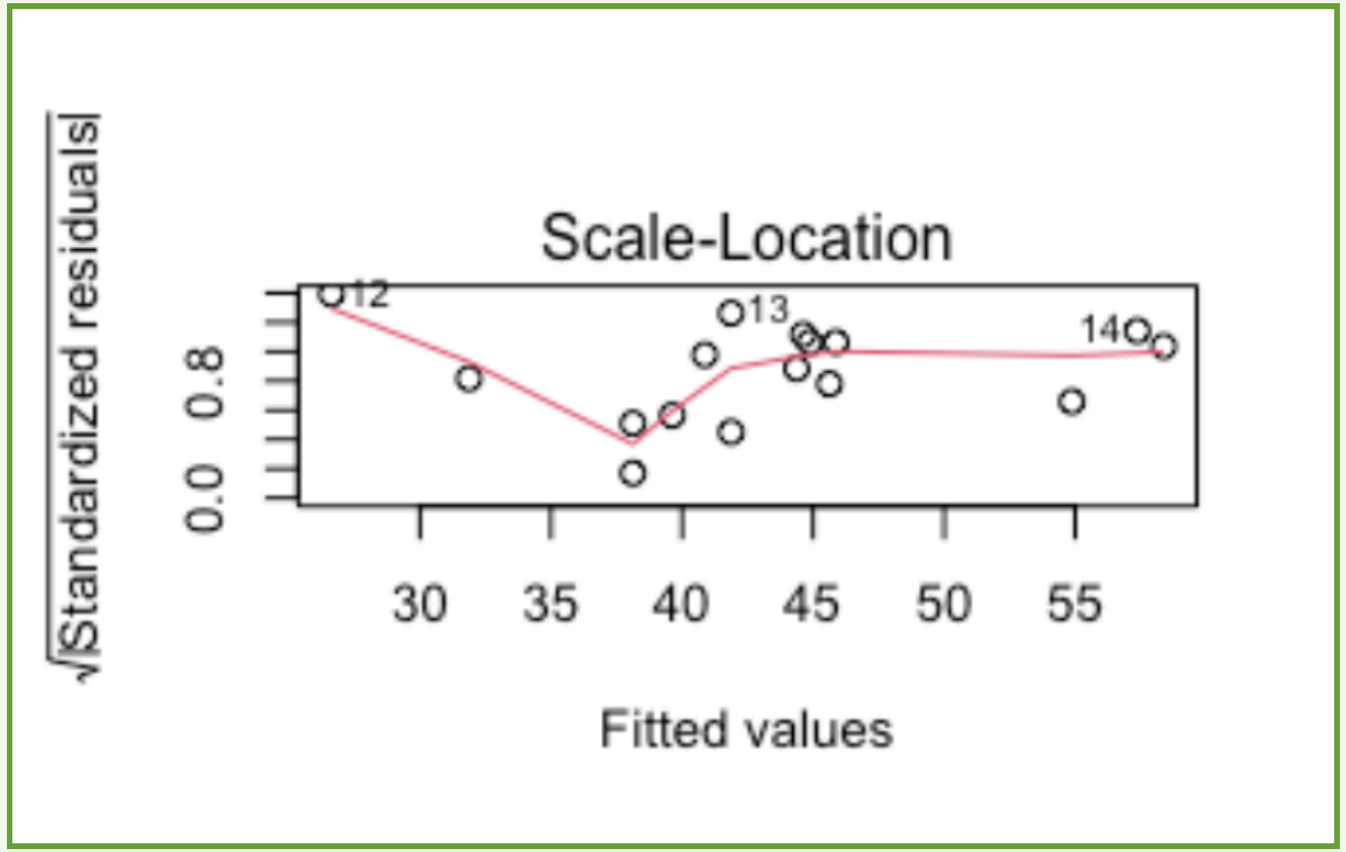
data:  yield by variety
Bartlett's K-squared = 2.6861, df = 3, p-value = 0.4426
```

Q-Q plot



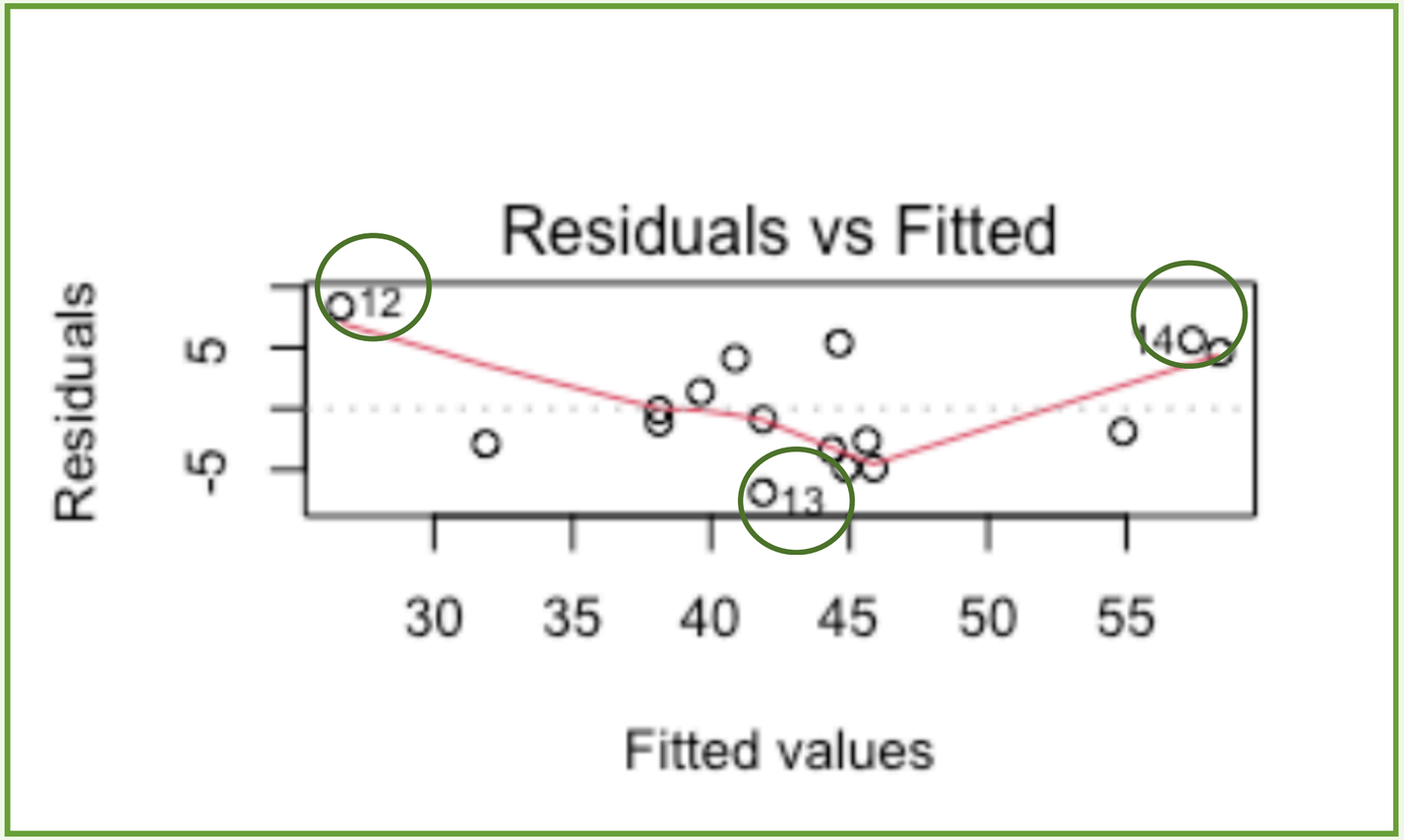
Most points follow the diagonal dotted line reasonably well throughout the middle, which is consistent with our Shapiro-Wilk result confirming normality was met. Points 13 at the lower left and 12 & 14 at the upper right deviate slightly from the line at the tails, but not severely enough to be concerning. This visually confirms that the normality assumption is satisfied for the melon dataset — a contrast to the clay dataset where points 15, 16, and 70 showed more severe tail deviations.

Scale-Location



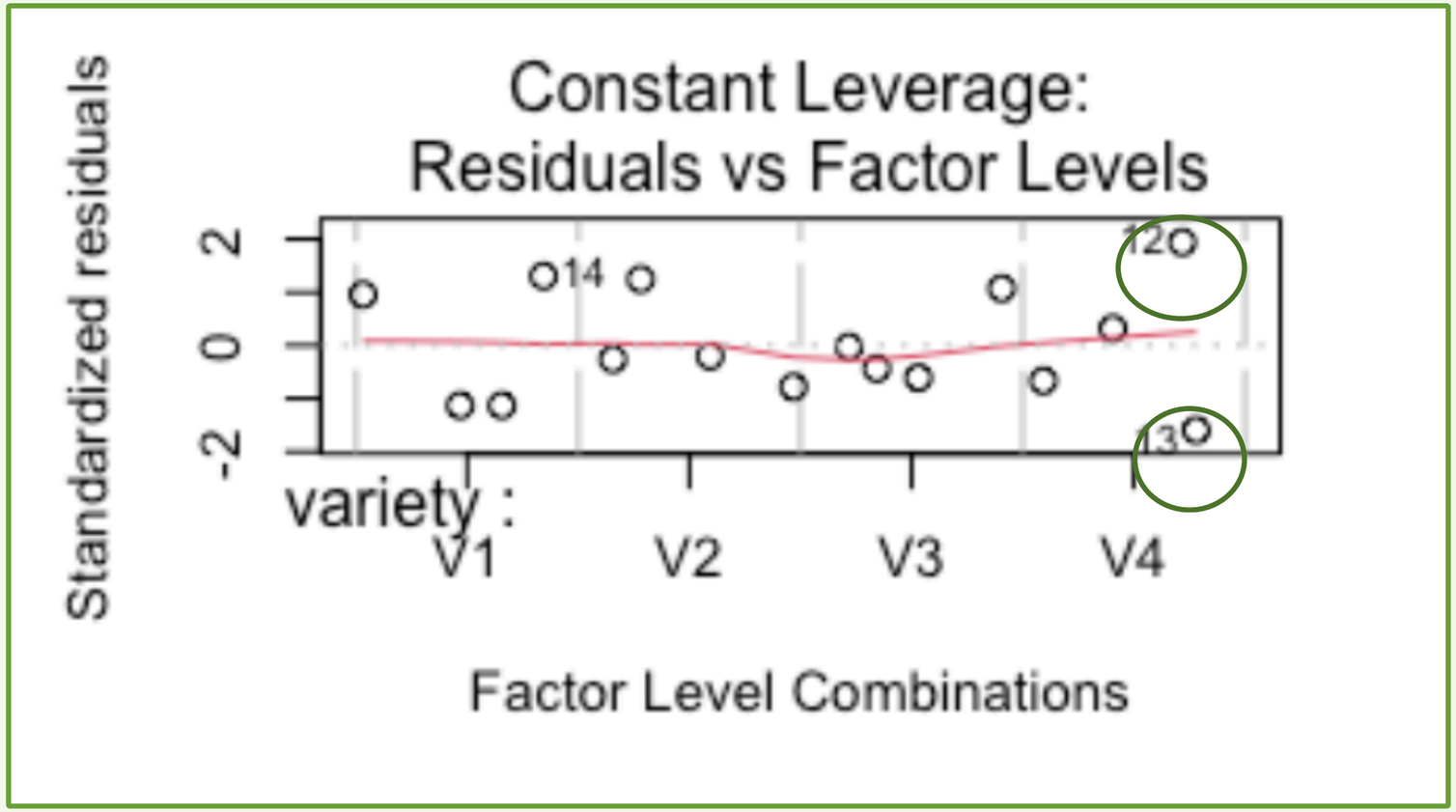
The red line drops from left to right and then levels off rather than being perfectly flat. Points 12 and 13 appear as the most unusual observations with higher standardized residuals. Despite this pattern the Bartlett test confirmed equal variances across varieties, so this uneven appearance is likely just an artifact of the very small sample size of 16 observations. With so few data points even one or two unusual values can visibly affect the shape of the line.

Residuals vs Fitted



The red line forms a V shape rather than being flat, indicating the model does not fit all values equally well across the range of fitted values. Points 12, 13, and 14 are flagged as the most unusual observations: point 12 has a large positive residual, point 13 has a large negative residual, and point 14 appears at the far right. With only 16 observations these few unusual points have a noticeable effect on the plot.

Residuals vs Factor Levels



This plot is specific to Latin Square designs and shows standardized residuals for each variety level rather than using leverage on the x-axis. Points 12 at the upper right of V4 and 13 at the lower right of V4 are again the most extreme observations across V1 and V4. The red line is relatively flat across all varieties, suggesting no single variety is consistently producing unusual residuals. Since the line stays close to zero across all varieties the model is not being driven by any one variety; no points appear severely influential.

Pairwise Comparison: TukeyHSD

Pairwise Comparisons

```{r}
TukeyHSD(model_melon, "variety")
```

```
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = yield ~ variety + row + col, data = melon)

$variety
        diff       lwr       upr     p adj
V2-V1  -5.00 -22.24715 12.247147 0.7535495
V3-V1   2.00 -15.24715 19.247147 0.9762029
V4-V1 -12.25 -29.49715  4.997147 0.1651663
V3-V2   7.00 -10.24715 24.247147 0.5398382
V4-V2  -7.25 -24.49715  9.997147 0.5142565
V4-V3 -14.25 -31.49715  2.997147 0.1017366
```

The Tukey HSD test compared all 6 possible pairs of melon varieties at α = 0.05.

Comparison	Difference	p adj
V2 vs V1	-5.00	0.754
V3 vs V1	+2.00	0.976
V4 vs V1	-12.25	0.165
V3 vs V2	+7.00	0.540
V4 vs V2	-7.25	0.514
V4 vs V3	-14.25	0.102

No pairwise comparisons were statistically significant — all adjusted p-values were well above 0.05. Although V4 shows the largest numerical differences from other varieties , 12.25 units below V1 and 14.25 units below V3, these differences did not reach significance. The confidence intervals for these comparisons are very wide, for example V4 vs V1 ranges from -29.50 to +5.00, which includes zero. When a confidence interval includes zero we cannot conclude a real difference exists.

The most likely reason nothing reached significance is the very small sample size. With only 4 replications per variety the test does not have enough statistical power to detect differences even when they appear large numerically. A larger experiment with more replications would be needed to draw stronger conclusions about variety differences. This result is consistent with the overall ANOVA which also found no significant variety effect (p = 0.102).

Concluding Dataset 2

1

No significant differences were found between the four melon varieties in terms of yield, meaning we cannot conclude any variety outperforms the others based on this experiment.

2

Both ANOVA assumptions were fully met, normality and equal variance, meaning the statistical analysis is reliable and no violations need to be noted, unlike the clay dataset.

3

The small sample size of only 16 observations and 4 replications per variety limited statistical power. A larger experiment with at least 8 to 10 replications per variety would be recommended to draw stronger conclusions.

COMPARING BOTH DATA SETS

	Clay	Melon
Design	RCBD	Latin Square
Blocking factors	1 (days)	(days)2 (row + col)
Treatment significant	Yes (p = 0.0015)	No (p = 0.102
Assumptions met	No	Yes
Post-hoc significant	Yes (32% clay)	No
Sample size	69	16
CV%	26.48%	16.22%

These two datasets tell different but equally valid scientific stories. Clay gave us clear significant results driven by a large sample and strong biological effect. Melon gave us non-significant results not because the experiment was wrong but because the sample size was too small to detect differences that may genuinely exist. Together they demonstrate that experimental design, sample size, and biological context all shape statistical outcomes.

Overall Conclusion:

These two datasets demonstrate that experimental design, sample size, and biological context all influence statistical outcomes. Clay showed strong significant treatment effects with assumption violations typical of field data, while melon showed no significant effects likely due to limited sample size. Together they highlight the importance of choosing appropriate designs and interpreting results in context.

Thank You!